AI Research Report

wildlife

General information about this document

Version	1.1.0
Version Date	07/04/2023
Modified by	Alexandre Saint-Albin
Creation Date	16/02/2023
Created by	Alexandre Saint-Albin

Modification History

Version	Date	Author	Comment
1.0.0	16/02/2023	Alexandre Saint-Albin	Document creation
1.1.0	07/04/2023	Alexandre Saint-Albin	Added AI results

Development of an AI for the classification of animal species as part of the Wildlife project

Introduction

The Wildlife project aims to develop a wildlife observation platform where users can share photos and videos of wild animals. The information generated by these posts, such as GPS coordinates, date, and species, will be used by professionals in the wildlife sector, such as zoologists and biologists. Artificial intelligence occupies a central place in this project, notably for the automatic recognition of animal species from images provided by users and connected cameras.

Classifying animal species from images represents a complex challenge, requiring high-performance image processing and machine learning algorithms. Convolutional Neural Networks (CNNs) have demonstrated their effectiveness in this field, particularly for image classification tasks. Transfer learning, a training technique, is commonly used to adapt a pre-trained model to new data, thus facilitating the creation of an animal species recognition model for the Wildlife project.

In this research report, we present the development of an AI intended for the classification of animal species as part of the Wildlife project, by exploring different CNN architectures and using transfer learning to optimize the performance of our model.

Dataset selection and usage

The choice of dataset is crucial to the success of any machine learning project. In our Wildlife project, we used the iNaturalist 2021 dataset, which is a collection of animal and plant species observations. iNaturalist is an online community and web application that allows users to document and share their nature observations, thus contributing to a vast set of data for biodiversity research.

We chose to use the iNaturalist dataset for several reasons:

- Richness and diversity of data: iNaturalist contains millions of observations of animal and plant species from all over the world, thus offering a rich and diverse data set to train our classification model.

- Accurate annotations: iNaturalist observations are annotated by experts and nature enthusiasts, which ensures high quality and precision of species labels.

- Consistency with the Wildlife project: iNaturalist shares similar goals with our project, encouraging users to observe and document wildlife. The use of the iNaturalist dataset is therefore consistent with the orientation and needs of our project.

There are other datasets that could be used to train an animal species classification model, such as ImageNet, which is a vast dataset containing millions of images divided into thousands of categories, including a large number of animal species. Another example is the CIFAR dataset, which includes 60,000 images of 10 different classes, some of which represent animals such as birds, cats, deer, and dogs. However, these datasets present certain limitations for our Wildlife project.

ImageNet, while being a rich and diverse dataset, covers a wide range of categories going far beyond animal species. Selecting and extracting a relevant subset of animal species from ImageNet could prove complex and tedious. Furthermore, ImageNet is a massive dataset that requires significant computational resources for training.

As for the CIFAR dataset, although it contains animal images, it is limited in terms of species diversity and only covers a small number of animals compared to iNaturalist. Additionally, CIFAR images are of low resolution (32x32 pixels), which could compromise the quality and precision of our animal species classification model.

In comparison, the iNaturalist dataset offers better coverage of animal species and superior image quality. Moreover, being specifically focused on biodiversity, it is better suited to our Wildlife project.
To adapt the iNaturalist dataset to our project, we decided to focus on 50 European animal species relevant to our study. This selection allowed us to focus on the most representative and relevant species for our wildlife observation platform. Furthermore, this reduction in the number of species was carried out for reasons of computational resource and time limitations. Training a model on a larger and more diverse dataset requires significant

computing power and longer processing time. By focusing on a reduced number of species, we were able to reduce computational and time requirements while ensuring we obtained a sufficiently accurate model for the selected species.

Deep learning algorithms

To develop our animal species classification model, we used the transfer learning method with three popular and efficient Convolutional Neural Network (CNN) architectures: **VGG16**, **ResNet50** and **InceptionV3**. Here is a detailed presentation of each algorithm, along with comparisons and justifications for their use in our project.

- VGG16: The VGG16 model, developed by the Visual Geometry Group research team at Oxford University, is a CNN architecture composed of 16 weighted layers. It was originally developed for the ImageNet image classification challenge in 2014 and achieved excellent results. VGG16 is appreciated for its simplicity and ease of implementation. However, its depth and large number of parameters lead to significant consumption of resources and computing time. In our project, we explored the use of VGG16 for animal species classification because it has been widely used and cited in previous work in the field of image classification.

- ResNet50: The ResNet50 model is a convolutional neural network with 50 weighted layers, developed by Microsoft Research. It is based on the ResNet (Residual Network) architecture which introduces residual connections, making it possible to resolve the problem of vanishing gradients in deep networks. ResNet won the ImageNet challenge in 2015 by setting a new record for accuracy. The main strength of ResNet lies in its ability to learn complex features while reducing convergence issues associated with deep networks. In our project, we used ResNet50 because of its proven performance and its ability to effectively handle deep networks for image classification.

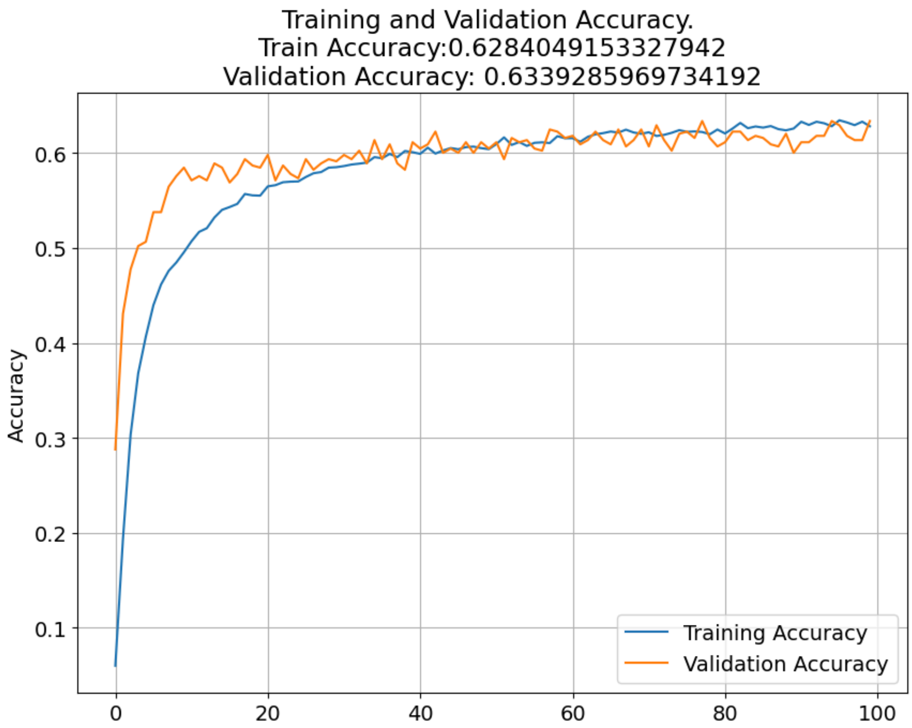
- InceptionV3: InceptionV3 is a CNN architecture developed by Google and is the third version of the Inception family. It uses Inception modules, which allow learning features at different spatial scales and reducing the number of parameters without sacrificing performance. InceptionV3 has achieved very good results in various image classification competitions, including ImageNet. The Inception architecture is particularly suited to our project due to its ability to process high-resolution images and its high performance with a relatively small number of parameters.

In our study, we chose to explore these three algorithms because they have proven themselves in previous work and image classification competitions. Furthermore, their use in other animal species classification projects provided us with reference points to evaluate their performance in the context of the Wildlife project.

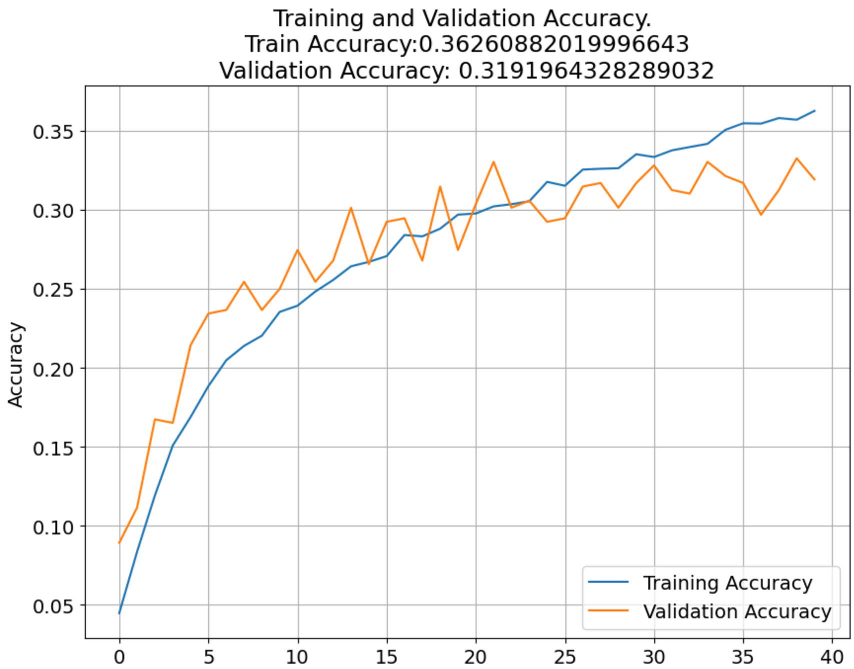
Results

After training and comparing the performance of these three architectures, we obtained the following results:

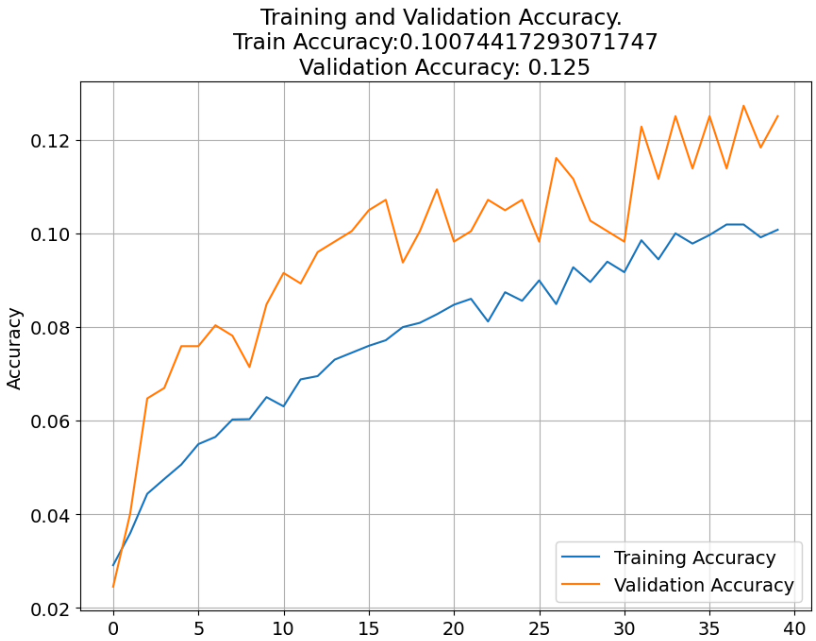
InceptionV3: 63% validation and 82% test



VGG16: 36% validation and 50% test



And ResNet50 with 12% validation and 4% test



(see notebooks for full results)

This superior performance is due to InceptionV3's ability to efficiently handle high-resolution images and learn complex features with a relatively low number of parameters, making it particularly suitable for animal species classification in our project.

Conclusion

During this project, we explored and implemented AI techniques for the classification of animal species. The challenges encountered included choosing a suitable model for image classification, selecting a relevant and representative dataset, and managing time and computational resource constraints.

We used transfer learning to train three popular deep learning models - VGG16, ResNet50 and InceptionV3 - and achieved the best results with the InceptionV3 architecture. Although our current model has some accuracy, there are several areas for improvement to further optimize performance and extend the capabilities of our animal species classification system.

Here are areas for improvement of the AI for the project:

Training on a larger number of species: Currently, our model is trained on a subset of 50 European animal species due to time and computational resource constraints. By gradually increasing the number of animal species considered, we could improve the generalization of our model and make it more useful for a larger number of users.

Enriching the dataset : Adding additional data, such as images taken in different lighting, weather or seasonal conditions, could improve the robustness and precision of the model by allowing it to better understand the variability of animal visual characteristics.

Exploration of data augmentation techniques: Using data augmentation techniques, such as rotation, scaling, cropping or flipping images, can help enrich the dataset and increase the model's ability to generalize from varied input data.

Optimization of hyperparameters: By experimenting and adjusting model hyperparameters, such as learning rate, batch size, or number of training epochs, we could potentially improve model performance.

Comparison with other deep learning architectures: Exploring other convolutional neural network architectures, as well as implementing newer and more advanced approaches to image classification, could allow us to find even more efficient models.

In summary, our Wildlife project allowed for the establishment of an animal species classification model based on the InceptionV3 architecture, providing a solid base for future improvements and extensions. By taking into account the areas for improvement mentioned above, we can continue to refine and optimize our model, thus contributing to a better understanding and better preservation of biodiversity.

Sources

iNaturalist: https://github.com/visipedia/inat_comp/tree/master/2021
CIFAR: https://www.cs.toronto.edu/~kriz/cifar.html
ImageNet : https://www.image-net.org/

InceptionV3 usages:
- https://hal.science/hal-03874232/document
- https://www.kaggle.com/c/dog-breed-identification
- https://towardsdatascience.com/car-classification-using-inception-v3-71b63e9825b7

ResNet50:
- https://medium.com/@nina95dan/simple-image-classification-with-resnet-50-334366e7311a
- https://www.kaggle.com/code/dskagglemt/car-image-classification-using-resnet50

VGG16:
- https://www.learndatasci.com/tutorials/hands-on-transfer-learning-keras/
- https://towardsdatascience.com/how-to-use-a-pre-trained-model-vgg-for-image-classification-8dd7c4a4a517